

Rushil Rawat

(413) 472-4816 | rushilrawat@umass.edu | rushilrawat.com | [linkedin/rushilrawat](https://linkedin.com/in/rushilrawat) | [github/rushilrawat](https://github.com/rushilrawat)

EDUCATION

University of Massachusetts, Amherst

Break Through Tech AI @ MIT · CodePath · Founders Circle · Dean's List

B.S. Computer Science + Business

Aug 2024 – May 2028

EXPERIENCE

AI/ML Fellow

May 2025 – May 2026

MIT Break Through Tech; Chambers Capital Ventures, Inc

Remote

- Engineered an **NLP segmentation pipeline** over 10,000+ survey responses, extracting TF-IDF, sentiment, and categorical features for K-means clustering
- Tuned K-means across k=2–8 using silhouette analysis, turning 8 behavioral clusters into 5 buyer personas
- Developed a **Streamlit** dashboard surfacing 11 purchase motivators, persona features to support startup market-positioning decisions

Data Analytics Extern

Jun 2025 – Sep 2025

Beats by Dre via Extern.com

Remote

- Analyzed 5,000+ customer responses with Python and **SQL**, engineering VADER/TextBlob sentiment and behavioral features across 6 audio market trends
- Implemented **scikit-learn** pipeline with standardized feature scaling and **K-means clustering**, uncovering 5 customer segments and 4 brand loyalty drivers via correlation analysis
- Built **SQL ETL** pipeline automating ingestion of 12 product metrics, designed real-time trend monitoring **Tableau** dashboard

Undergraduate Researcher

Dec 2024 – Feb 2025

Wireless Sensors and System Lab, UMass Amherst

Hybrid

- Fine-tuned a ResNet CNN with **TensorFlow & OpenCV** for UAV fire detection, achieving 0.81 accuracy, 0.78 precision, 0.76 recall, and 0.77 F1
- Automated annotation and dataset management for 2,000+ images, using LabelImg expanding dataset by 40%
- Integrated Model inference into GIS dashboard, tested model in low-visibility for emergency robustness

PROJECTS

HarmScope | Python, LLM/RAG, FAISS, DuckDB, NLP, statsmodels

Aug 2026 – Present

- Engineered NLP pipeline over 16.5M CFPB complaints & 3.8M narratives, using sentence embeddings, FAISS **vector search** and **unsupervised** clustering to identify emerging consumer harm patterns
- Implemented time-series anomaly detection with a leakage-checked evaluation harness, backtesting signals against 112 enforcement actions and reducing false alerts from 10.7% to 0.45%
- Built a grounded **LLM/RAG** layer with hybrid retrieval, structured outputs, and blinded LLM evaluation harness

CapitolWatch Insider Trading | Python, FastAPI, PostgreSQL, React, Docker

Jul 2026

- Built a **data pipeline** pulling 72,000 congressional stock trades from scanned PDF filings, using OCR and fuzzy name-matching in **PostgreSQL**
- Analyzed trade timing of 350+ members by permutation and false-discovery-rate correction, flagging 4 portfolios
- Automated** agent for daily data update; **cached** unchanged filings to save memory, FastAPI backend

FDA Medical Device Risk Analyzer | Python, Pandas, scikit-learn, Tableau

Dec 2025 – Present

- Implemented data pipeline processing **500,000+** FDA adverse event records, applying TF-IDF and K-means clustering to identify recurring device-failure patterns by frequency and severity
- Built an **LLM-assisted RAG** layer using sentence **embeddings** and FAISS retrieval, generating failure summaries with source-linked evidence from narratives

SKILLS

Languages: Python, SQL, R, TypeScript/JavaScript, Java, C++

AI & Machine Learning: TensorFlow, scikit-learn, NLP, LLMs, RAG, Embeddings, Vector Search (FAISS), OpenCV, Anomaly Detection, Model Evaluation

Backend & ML Systems: FastAPI, REST APIs, Docker, Git, CI/CD, Linux/Unix, AWS, PostgreSQL, DuckDB, MySQL

Data & Statistics: Pandas, NumPy, SciPy, statsmodels, ETL/Data Pipelines, Streamlit, Tableau, Plotly Matplotlib